

Generation Interconnection System Impact Study Report

For Queue Project Q17-0768

Nambe 138 kV

May 02, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0768. The Interconnection Customer proposes a 105.1 MW solar photovoltaic facility interconnecting at the Nambe 138 kV substation in the Tennessee area. The request was received on March 30, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 228 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,105,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The McClain Energy Facility - Coal 138kV is overloaded to 125.5% of its 254 MVA emergency rating for the loss of the Manatee - Hintonap 345kV Line. The project contributes 6.0% DFAX (6.3 MW) to this overload.

Red Hills - Beardsley St 138kV: 124.4% loading (355 MVA) following the outage of the Lowmoor - Hinton Ap 345kV; DFAX = 8.0%.

De Kalb - Oxnard 138kV: 110.0% loading (223 MVA) following the outage of the Lowmoor - Hinton Ap 345kV; DFAX = 5.2%.

The Oxard - Hinton Ap 138kV is overloaded to 109.9% of its 229 MVA emergency rating for the loss of the Lowmoor - Hinton Ap 345kV line. The project contributes 5.2% DFAX (5.4 MW) to this overload.

Vaseaux Lake - Red Hills 138kV: 108.7% loading (415 MVA) following the outage of the Lowmoor - Hinton Ap 345kV; DFAX = 8.0%.

McClain Energy Facility - De Kalb 138kV: 108.6% loading (220 MVA) following the outage of the Hinton Ap - Lowmoor 345kV; DFAX = 5.2%.

The McClain Energy Facility - Vaseaux Lake 138kV Line is overloaded to 108.6% of its 418 MVA emergency rating for the loss of the Lowmoor - Hinton Ap 345kV. The project contributes 8.0% DFAX (8.4 MW) to this overload.

For the loss of Aspen - Lowmoor 345kV, the Lowmoor 345/138 kV Autotransformer loads to 103.3% of its 449 MVA rating. Project DFAX: 7.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcclain Energy Facility - Coal Substation 138kV Line	Rebuild line with higher-capacity conductor	\$6,003,000	9% (\$567,000)
Beardsley St - Red Hills 138kV	Rebuild line with higher-capacity conductor	\$9,783,000	8% (\$800,000)
De Kalb - Oxnard 138kV	Reconductor line	\$1,888,000	10% (\$190,000)
Oxard - Hinton Ap 138kV	Reconductor line	\$1,321,000	10% (\$133,000)
Red Hills - Vaseaux Lake 138kV	Reconductor line	\$6,632,000	8% (\$542,000)
Mcclain Energy Facility - De Kalb 138kV	Reconductor line	\$1,920,000	10% (\$193,000)
Mcclain Energy Facility - Vaseaux Lake 138kV Line	Reconductor line	\$4,723,000	8% (\$386,000)
Lowmoor 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$15,775,000	8% (\$1,294,000)
	Total allocated to project		\$4,105,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.